

NOVA GEAR SALES ANALYSIS

SQL & Power BI Data Analytics Project

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Tools:

MySQL • Power BI • Excel

EXECUTIVE SUMMARY

This project analyses Nova Gear sales data to evaluate sales performance, revenue drivers, order outcomes, and sales trends. The dataset was cleaned and validated using SQL in MySQL, including duplicate removal, date standardization, text-field validation, and data-quality checks. The cleaned dataset was then analysed using SQL to generate key business metrics and insights and was connected to Power BI to develop an interactive dashboard.

The final analysis provides a consolidated view of revenue, orders, customers, product categories, countries, payment methods, order status, and monthly sales trends. Interactive slicers allow users to explore the results by country, product category, and order status.

1 BUSINESS PROBLEM

Nova Gear requires a clear view of its sales performance to understand where revenue is being generated, how orders are performing, and where potential revenue leakage is occurring. The analysis was designed to transform the raw sales data into reliable information that can support business decision-making.

The project focused on answering the following key business questions:

- What is the overall revenue and order performance?
- Which product categories generate the most revenue?
- Which countries contribute the most to revenue?
- Which payment methods generate the highest revenue?
- How does revenue change over time?
- What proportion of revenue comes from completed, refunded, and cancelled

orders?

- How can the findings be presented in an interactive dashboard for easier decision-making?

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DATASET OVERVIEW

The Nova Gear Sales dataset contains transaction-level sales records covering multiple years of sales activity. The original dataset contained **9,869 records**. After data cleaning and removal of exact duplicate records, the cleaned dataset contained **9,721 records**.

The dataset includes information about customers, locations, products, pricing, quantities, devices, payment methods, order dates, and order status. Key fields used in the analysis included:

- Customer name
- Country and city
- Product category and product name
- Unit price and quantity
- Device type
- Payment method
- Order status
- Order date

The cleaned dataset was used as the primary source for the SQL analysis and Power BI dashboard.

3 DATA CLEANING & AND PREPARATION

3.1 DATA QUALITY ASSESSMENT

The raw Nova Gear Sales dataset was first inspected in MySQL to assess its structure, data types, missing values, duplicate records, and consistency of key fields. This initial assessment helped identify the cleaning requirements before the data was used for analysis.

The main data-quality checks included:

- Reviewing the table structure and column data types
- Checking for missing and blank values
- Checking for duplicate records and duplicate order IDs
- Reviewing text fields for unnecessary spaces and blank values
- Examining the order-date field for inconsistent date formats
- Validating the cleaned dataset after the transformation process

3.2 DATA CLEANING AND TRANSFORMATION

The identified data-quality issues were addressed using MySQL. Exact duplicate records were removed by creating a distinct cleaned table, reducing the dataset from 9,869 records to 9,721 records.

The order data field was converted from a text field containing mixed date formats into a standardized DATE format. The transformed dates were validated to ensure that no invalid or unconverted dates remained.

Text fields were also reviewed for unnecessary spaces, blank values, and consistency. Validation checks using trimming and length checks confirmed that the relevant text fields did not contain unintended blank values.

A cleaned table named Nova_Gear_Sales_Clean was created and used as the primary dataset for the subsequent SQL analysis and Power BI dashboard.

3.3 DATA VALIDATION

After the cleaning and transformation process, the cleaned dataset was validated in MySQL to confirm that the expected records, fields, and key analytical values were preserved. The final cleaned dataset contained 9,721 records.

Additional validation confirmed 1,061 distinct customer names and 9,681 distinct order IDs. These figures were used as reference values for the Power BI dashboard to maintain consistency between the SQL analysis and the visualization layer.

4 SQL ANALYSIS AND KEY FINDINGS

4.1 KEY BUSSINESS METRICS

SQL analysis was performed on the cleaned nova_gear_sales_clean table to calculate the core business metrics used in the project. These metrics provided the numerical foundation for the Power BI dashboard.

The key metrics were:

- **Total Orders:** 9,681 unique orders
- **Total Customers:** 1,061 distinct customer names
- **Total Revenue:** approximately \$5.12 million
- **Completed Revenue:** approximately \$4.75 million
- **Leakage Rate:** 7.10%

These metrics were used to provide an overall view of sales performance and to support further analysis of product categories, countries, payment methods, order status, and monthly revenue trends.

4.2 REVENUE BY PRODUCT CATEGORY

Revenue was analysed across the different product categories to identify the major contributors to overall sales performance.

The analysis showed that **Computing** generated the highest revenue, followed closely by **Phones & Tablets**. **Audio** was the next-largest contributor, while **Smart Home, Wearables, and Accessories** generated comparatively lower revenue.

This analysis helped identify the product categories that contributed most to total revenue and provided a basis for comparing category-level sales performance in the Power BI dashboard.

4.3 REVENUE BY COUNTRY

Revenue was analysed by country to identify the markets contributing most to overall sales. The analysis showed that the **United States generated the highest revenue**, followed by **China, Nigeria, Germany, France, Canada, Kenya, and Australia**.

The country-level analysis provided a clearer view of the geographical distribution of revenue and was incorporated into the Power BI dashboard for interactive comparison. Revenue was analysed by country to identify the markets contributing most to overall sales.

The analysis showed that the **United States generated the highest revenue**, followed by **China, Nigeria, Germany, France, Canada, Kenya, and Australia**.

The country-level analysis provided a clearer view of the geographical distribution of revenue and was incorporated into the Power BI dashboard for interactive comparison.

4.4 REVENUE BY PAYMENT METHOD

Revenue was analysed by payment method to identify the payment channels contributing to sales performance.

The analysis showed that **Mobile Money generated the highest revenue**, followed by **Card and Apple Pay**. Other payment methods, including **WeChat Pay, Bank Transfer, and PayPal**, contributed additional revenue, while some methods generated comparatively lower amounts.

This analysis provided insight into customer payment preferences and was presented in the Power BI dashboard for interactive comparison.

4.5 REVENUE BY MONTH

Revenue was analysed by month to identify changes in sales performance over time and to observe overall revenue trends.

The monthly analysis provided a view of how revenue changed across the period covered by the dataset. This helped identify periods of higher and lower sales activity and supported trend analysis in the Power BI dashboard.

The monthly revenue trend was presented in the Power BI dashboard to allow users to explore sales performance over time.

4.6 REVENUE BY ORDER STATUS

Revenue was analysed by order status to understand the distribution of completed, refunded, and cancelled orders.

The analysis showed that completed orders accounted for the largest share of revenue, while refunded and cancelled orders represented revenue that did not result in completed sales.

This analysis supported the assessment of revenue leakage and was presented in the Power BI dashboard using an order-status visual for interactive comparison.

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POWER BI DASHBOARD

The cleaned Nova Gear Sales dataset was connected to Power BI to develop an interactive dashboard for visual analysis of sales performance. The dashboard was designed to present the key business metrics and SQL findings in a clear and accessible format.

The dashboard allows users to explore revenue performance across product categories, countries, payment methods, order status, and monthly trends.

5.1 DASHBOARD DESIGN AND KPIs

. The dashboard was structured around five key performance indicators: Total Orders, Total Customers, Total Revenue, Completed Revenue, and Leakage Rate.

These KPIs provide a high-level view of sales volume, customer reach, revenue generated, completed sales value, and revenue associated with non-completed orders.

The KPI values were aligned with the validated SQL results to maintain consistency between the analytical calculations and the Power BI dashboard.

5.2 DASHBOARD VISUALS

The dashboard includes five visualisations designed to present the main findings from the SQL analysis.

The visuals show revenue by product category, revenue by payment method, revenue by country, revenue by month, and revenue by order status.

These visuals provide different views of sales performance and allow users to compare revenue across categories, geographical markets, payment channels, order outcomes, and time periods.

5.3 INTERACTIVE FILTERS

Interactive slicers were included to allow users to filter the dashboard by country, product category, and order status.

These filters enable users to examine specific segments of the sales data without changing the underlying analysis. The interactive design makes it easier to compare sales performance across different markets, product categories, and order outcomes.

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KEY FINDINGS

The analysis identified several important patterns in Nova Gear's sales performance. Revenue was concentrated in a small number of product categories and geographical markets, while payment methods showed differences in revenue contribution.

The analysis also showed that completed orders generated the majority of revenue, while refunded and cancelled orders accounted for a smaller share. Monthly revenue analysis provided an overview of changes in sales activity over the period covered by the dataset.

These findings were incorporated into the Power BI dashboard to provide an interactive view of sales performance and support further exploration of the underlying data.

Based on the analysis, the following recommendations are proposed:

- Monitor product-category performance regularly to understand changes in revenue contribution and identify areas requiring further attention.
- Benchmark the sales performance of higher-revenue countries to identify practices that may be adaptable to lower-revenue markets. These may include effective payment channels, product-category focus, customer engagement approaches, and market-specific sales strategies. Such practices should be tested and evaluated before wider implementation.
- Review sales performance across countries to understand geographical differences and support informed market decisions.
- Monitor refunded and cancelled orders to identify potential sources of revenue leakage and improve completed sales performance.
- Track payment-method performance to understand customer payment patterns and support effective payment-channel management.
- Continue using the Power BI dashboard to monitor key performance indicators and explore changes in sales performance over time.

This project demonstrated how SQL can be used to clean, validate, analyse, and transform sales data into meaningful business insights. The Nova Gear Sales dataset was successfully prepared in MySQL, with duplicate records removed, dates standardised, and key data-quality checks completed.

The analysis identified important patterns in revenue performance across product categories, countries, payment methods, order status, and time periods. These findings were then presented through an interactive Power BI dashboard, allowing users to explore sales performance and key business metrics.

Overall, the project shows how SQL, Excel, and Power BI can be combined to support data-driven business analysis and decision-making. The resulting analysis provides a practical framework for monitoring sales performance, identifying revenue leakage, and exploring opportunities for improvement.

